

Review

Evidence of sensorimotor deficits in functional ankle instability: A systematic review with meta-analysis

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Abstract

Functional ankle instability (FAI) has been associated with impaired sensorimotor function; however individual studies have produced conflicting results. In an attempt to reduce this ambiguity, a systematic review with meta-analysis was undertaken to determine which sensorimotor deficits exist with FAI. Fifty-three studies assessing sensorimotor factors in subjects with FAI were included from 465 identified articles. Studies were rated for methodological quality and data were pooled for peroneal reaction time, joint position sense, and postural sway during single-leg stance and time to stabilisation from a single-leg jump. Data on joint movement sense were unable to be pooled. When subjects with unstable ankles were compared to healthy controls, sensorimotor impairments were demonstrated for passive joint position sense (mean difference (MD) = 0.7°, 95% confidence interval (CI): 0.2–1.2°, $p = 0.004$), active joint position sense (MD = 0.6°, 95% CI: 0.2–1.0°, $p = 0.002$), postural sway in single-leg stance (standardised MD (SMD) = 0.6, 95% CI: 0.2–1.0, $p = 0.002$), the star excursion balance test (SMD = 0.4, 95% CI: 0.1–0.7, $p = 0.009$), and time to stabilisation from a single-leg jump in a medio-lateral (MD = 0.6 ms, 95% CI: 0.4–0.8, $p < 0.0001$) and an antero-posterior direction (MD = 0.7 ms, 95% CI: 0.4–1.0, $p < 0.0001$). Peroneal reaction time was not affected. Sensorimotor deficits occur for joint position sense and postural control in subjects with FAI. Deficits in peroneal muscle reaction time following perturbation are not evident.

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Keywords: Ankle injuries; Proprioception; Reaction time; Joint position sense; Kinesthesia; Postural sway

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1. Introduction

Following lateral ankle sprain injury there is a high risk of recurrence (>70%),¹ which may result in the development of recalcitrant conditions including chronic instability.² Chronic ankle instability can include both mechanical and functional

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ankle instability (FAI). Mechanical instability refers to laxity of a joint due to loss of mechanical restraint such as ligamentous tissues,³ while FAI describes the perception that the ankle gives way, is weaker, more painful, or less functional following injury.⁴

In an attempt to minimise symptoms of instability and injury recurrence following ankle sprain rehabilitation traditionally focuses on identifying and correcting factors or impairments, thought to be associated with ankle instability.⁵ Numerous published studies have identified various sensorimotor impairments associated with FAI. Specifically these include; impaired balance,^{6–8} increased peroneal reaction time,⁹ and decreased joint movement sense.^{10,11} However, studies investigating the occurrence of sensorimotor impairments associated with ankle instability have produced inconsistent findings.^{6–9,12–14} Because of such inconsistencies, sensorimotor factors significantly associated with FAI are not clearly identified in the literature, limiting the underlying basis for rehabilitation following ankle injury.

Several narrative reviews have provided an overview of sensorimotor factors associated with ankle instability (e.g.^{5,15}), however, narrative reviews are potentially biased,¹⁶ and they may not provide unambiguous evidence of factors associated with symptoms of ankle instability. By performing a systematic review, methodology is transparent and the quality of individual studies can be rated, minimising potential sources of bias¹⁷ and allowing improved interpretation of factors associated with ankle instability.

The aim of this review was to identify key sensorimotor factors associated with symptoms of FAI by systematically identifying and assessing the methodological quality of relevant literature. Findings of individual studies were statistically combined in a meta-analysis, where possible, to derive pooled estimates for the differences between unstable and healthy ankles for specific sensorimotor factors.

2. Methods

Electronic database searches were performed to identify all relevant studies of sensorimotor factors associated with ankle instability subsequent to lateral ligament injury. Standardised search terms without language restrictions (combinations of key words for ankle, injury, instability and sensorimotor factors including balance, proprioception, kinesthesia, motor control and reaction time) were entered into MEDLINE, CINAHL, EMBASE and SPORTDiscus databases from their earliest available date to November 2006. Supplementary database searches of Scopus and PubMed were performed using similar keywords or MeSH headings relevant to the databases, and reference lists of included papers were manually checked.

Papers returned from electronic searching that were not original research, full papers in a peer reviewed journal, or where the source population was inappropriate (for example, neurological disorders, anterior cruciate ligament injury,

diabetics) were not included. Two reviewers, blinded to authors, journal and institution, independently reviewed titled abstracts retrieved from the initial search. Papers were included if they: assessed subjects with FAI persisting longer than six weeks following ankle injury; compared FAI to controls, or in the case of unilateral instability to the contralateral limb; and if at least one of the following sensorimotor measures were assessed; (i) postural control, (ii) joint position sense, (iii) kinesthesia, (iv) muscle reaction time. Intervention studies were included if baseline comparison of sensorimotor measures were recorded. Where information from the title and abstract was inadequate to determine initial inclusion, full papers were retrieved and assessed. Uncertainty in selection status of individual papers was resolved by mutual consensus among the reviewers.

Study quality was independently assessed by two of the three members of the research team blinded to the authors, journal and institution using a quality index for non-randomised studies.^{18,19} This index has previously demonstrated high internal consistency and inter-rater reliability.¹⁸ An adapted 15 item version of the original index was previously used in a review of intrinsic and extrinsic risk factors associated with heel pain.¹⁹ The adapted version was deemed appropriate for this review as it does not apply questions relating to methodological design validity associated with an intervention. However, unlike the adaptation used by Irving et al.,¹⁹ we maintained an additional item related to blinding of the observer to the sensorimotor measure, as we believed that this was possible for some measures. The quality index used in this current review therefore assessed 16 items of study quality ([Supplementary data](#)) scored as one of four response categories. Any ambiguities in scores were resolved by mutual consent or by a third reviewer where necessary. In studies where there was no control group (i.e. within subject limb comparisons), items relating to subject and control recruitment were not applied. For the purposes of this study the quality of studies meeting 60–74% of the applicable criteria was considered as moderate; >75% as high and <60% as low.

Information on study and sample characteristics was extracted and is presented in [Table S1](#). Where one aspect of sensorimotor function was assessed with multiple measures reported (e.g. kinesthesia for different directions and at different test speeds¹¹), decision rules formulated and agreed upon by all three members of the research team were applied to prioritise the data to be extracted. Measures occurring most frequently across individual studies were preferred: total or combined scores were preferred over single scores; measures in the frontal plane were preferred over other directions; and where measures were not consistent with other studies, the condition considered most challenging was preferred.

For continuous outcomes we calculated mean and 95% confidence intervals (CI) for differences between groups (where a control group was included) and the difference between limbs for unilateral instability (in the absence of a control group). Where studies were similar in terms of the

sensorimotor factor measured (e.g. peroneal reaction time, active joint position sense) they were grouped accordingly. Pooled estimates of the effects of ankle instability on sensorimotor measures were obtained using a random effects model (RevMan 4.2²⁰) if three or more homogeneous studies were available. Where sensorimotor measures quantifying the same construct used incompatible measurement scales (e.g. postural sway) pooled estimates were calculated as standardised mean differences (SMD).

3. Results

The search strategy identified 465 papers for independent screening with 53 meeting the criteria for inclusion in the systematic review (Fig. 1). Originally 56 studies were included but, upon revealing authors, three of these appeared to be duplicated papers in different languages or repeated data. Of the 53 studies, 13 performed interventions (such as taping, surgery or exercise) but used repeated measures of sensorimotor outcomes. Overall, 26 studies performed between group comparisons only, 11 compared limbs in subjects with unilateral instability only and one study looked at correlated instability scores and sensorimotor measures in healthy and unstable subjects. Twenty-nine studies assessed postural control, 11 muscle response time, 13 joint position sense (JPS) and six kinesthesia (passive joint movement detection). Most studies involved young, physically active participants. Details

of the included studies, identifying participant characteristics and sensorimotor outcomes, are shown in Table S1.

Quality assessment scores from the included studies are presented in Supplementary data Table S2. Twenty-seven studies (51%) met the criteria completely or partially in 60% or more of the applicable item categories. Ten studies (19%) met less than half of the methodological quality criteria items, with the two most common weaknesses occurring in reporting of study methodology with lack of observer blinding (item 15), and inadequate information to determine if the control and source population were representative (items 11, 12, 21, 22 and Table S2).

Eleven studies assessed muscle reaction time following perturbation. All of these assessed the peroneal muscle group, and two additionally assessed tibialis anterior. Data for peroneal muscle reaction time to inversion perturbation from individual studies is provided in Table 1. Data was pooled from 6 studies^{9,21–25} for between group comparisons, however only two of these studies achieved $\geq 60\%$ of the methodological quality criteria.^{23,25} There was no difference in reaction time between subjects with FAI and uninjured controls (mean difference (MD) = 7.8 ms, 95% CI: -1.4 to 17.1 , $p = 0.10$) (Fig. 2). One study comparing tibialis anterior reaction time in FAI with healthy controls⁹ found delayed muscle reaction in unstable subjects (Table 1). Comparisons between limbs in subjects with unilateral ankle instability did not show differences for peroneal muscle reaction (pooled estimate of MD = 4.5 ms, 95% CI: -2.0 to 11.0 , $p = 0.17$),^{9,13,26–29} or

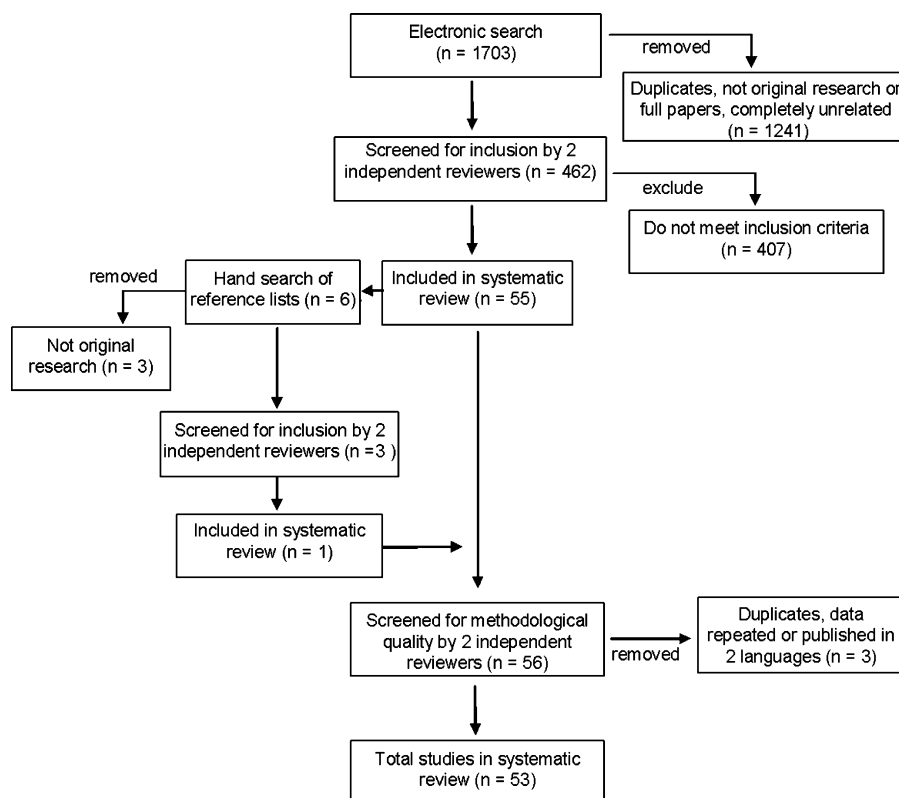


Fig. 1. Process for selection of studies to be included in the systematic review.

Table 1

Differences in muscle reaction time between unstable and healthy ankles from individual studies.

Muscle Group	Study	Reaction time (ms)	Comparison	Mean difference (95% CI)	% of QC met
Peroneal	23	Peroneal reaction to inv.	BG	11.2 (2.5–19.9)	63
	22	Longus (with out brace/tape) (ms) to inv	BG	–9 (–13.7 to –4.4)	56
	29	Longus reaction to inv.	BL	2.2 (–1.3 to 5.4)	67
	21	Peroneal reaction to inv.	BG	11.0 (4.1–18.0)	56
	26	Longus reaction to inv.	BL	10.8 (6.8–14.9)	62
	25	Longus reaction to inv.	BG	1.4 (–3.3 to 6.1)	63
	28	Longus reaction to inv.	BL	0.7 (–2.6 to 4.0)	56
	24	Longus reaction to inv.	BG	17 (13.8–20.2)	31
	9	Longus reaction to inv. (ipsilateral limb)	BG BL	15.8 (10.3–21.3) 0.3 (–6.3 to 6.9)	44
	27	Longus reaction to inv. (without tape)	BL	15.7 (13.1–18.3)	54
	13	Peroneal reaction to combined inv. PF	BL	–6.7 (–17.7 to 4.3)	54
Tibialis anterior	9	Reaction to inv. (ipsilateral limb)	BG BL	19.1 (14.2–24.0) 2.5 (–3.9 to 8.9)	44
	13	Reaction to combined inv. PF	BL	–3.7 (–14.5 to 7.1)	54

QC = quality criteria; BG = between group; BL = between limb; PF = plantar flexion; inv. = inversion.

from individual studies for tibialis anterior reaction time,^{9,13} (Table 1).

Six studies assessed passive movement detection (PMD).^{10,11,30–33} Four of these reported continuous data, with three examining movements in the frontal plane^{11,31–33} and one in the sagittal plane.³² The remaining two studies reported dichotomous outcomes of movement sense into plantar flexion^{10,30} (Table 2). For PMD in the frontal plane, one study performed comparisons to a control group¹¹ and two compared limbs of subjects with unilateral instability,^{31,33} therefore data could not be pooled. In two of the three studies, movement detection was reduced in subjects with FAI (Table 2), and these papers were of high quality ($\geq 75\%$ of criteria met, Table S2). Passive movement detection for combined plantar flexion and dorsiflexion was not impaired in subjects with FAI³² (Table 2). For passive plantar flexion using dichotomous outcomes,^{10,30} significantly reduced movement detection was reported in the injured compared to the contralateral ankle in subjects with unilateral instability (Table 2). Studies for PMD in the sagittal plane were generally of low quality (53–61% of criteria met).

Thirteen studies reported data for JPS (11 included active and six included passive positioning movements,

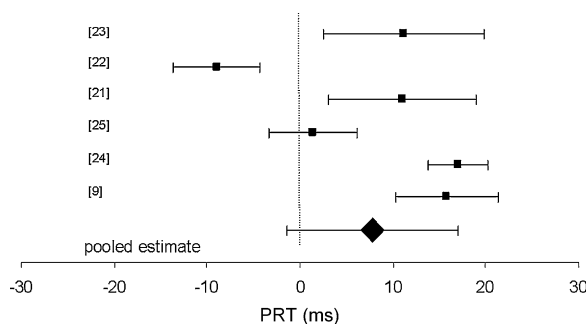


Fig. 2. Peroneal reaction time (PRT) to inversion perturbation. Distribution of calculated estimates from 6 studies for differences between subjects with unstable ankles compared to healthy control subjects (■). Positive values favor healthy control ankles. Pooled estimate (◆) (random effects model) based on all 6 studies. Horizontal bars represent 95% confidence intervals.

Table 3). For passive JPS, pooled data from six studies^{34–39} (Table 3) showed that subjects with instability had reduced JPS (MD = 0.7°, 95% CI: 0.2–1.2°, $p = 0.004$) compared to healthy control subjects (Fig. 3). Study quality was mixed: one study was of high quality and three studies met less than 60% of applicable quality criteria (Table 3). Similarly, for active JPS, pooled data from 10 studies,^{34,37–45} (study 11⁴⁶ was unsuitable for pooling, Table 3) found a mean deficit of 0.6° (95% CI: 0.2–1.0°, $p = 0.002$) for subjects with FAI compared to controls (Fig. 3). Overall methodology of studies on active JPS was limited with seven of 11 not meeting 60% or more of quality criteria (Table 3). The results from pooled data of four studies for active position sense^{37,38,40,44} (Table 3), using between limb differences in unilaterally unstable subjects (MD = 0.5°, 95% CI: 0.3–0.7°, $p < 0.0001$),

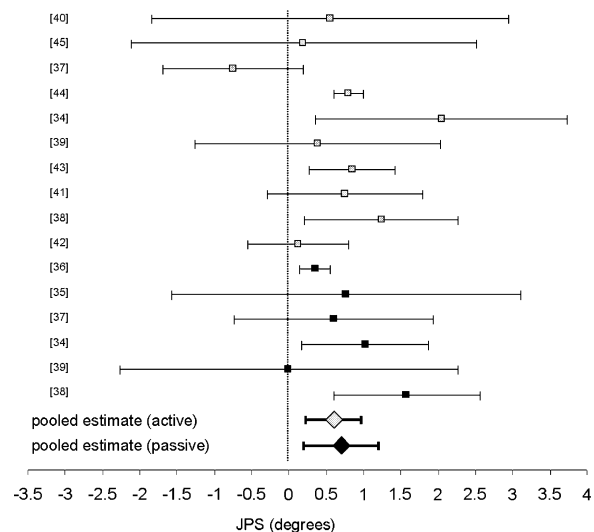


Fig. 3. Joint position sense (JPS) (degrees). Distribution of calculated estimates from 10 studies on active JPS (■) and 6 studies on passive JPS (■) for differences between subjects with unstable ankles compared to healthy control subjects. Positive values favor healthy control ankles. Pooled estimates (random effects model) for active JPS (◆) based on all 10 studies and passive JPS (◆) based on all 6 studies. Horizontal bars represent 95% confidence intervals.

Table 2

Differences in kinesthesia (passive movement detection [PMD]) between unstable and healthy ankles from individual studies.

	Study	PMD direction (°)	Comparison	Mean difference (95% CI) (or reported finding*)	% of QC met
Frontal plane	11	Combined inv. and ev. (0.1°/s)	BG	0.6° (0.2–1.0°)	75
	31	Inv. (0.5°/s)	BL	0.34° (–1.62° to 2.30°)	77
	33	Inv.	BL	1.1° (0.0–2.2°)	77
Sagittal plane	32	Combined DF and PF (0.1°/s) (for 70% detection)	BG	0.2° (–0.1° to 0.4°)	56
	30	PF (0.3°/s) for (yes/no response)	BL	*Significantly greater difficulty detecting passive movement for movement ($p < 0.05$) but not non-movement conditions	61
	10	PF (0.3°/s) for (yes/no response)	BL	*Significantly greater difficulty detecting movement for movement ($p < 0.01$) and non-movement conditions ($p < 0.05$)	53

QC = quality criteria; BG = between group; BL = between limb; inv. = inversion; DF = dorsiflexion; PF = plantar flexion.

* Original finding reported as appropriate data could not be extracted to estimate effect.

were similar to group comparison findings. Passive JPS was assessed between limbs in two low quality studies^{37,38} that yielded conflicting results (Table 3).

Twenty-nine studies reported sensorimotor measures of postural control (Table 4). Specifically, these included time to stabilisation from a 1-legged jump, Star Excursion Balance Test (SEBT), Balance Error Scoring System (BESS) and postural sway velocity and displacement area in single-leg stance under stable or challenging conditions (e.g. eyes closed, moving or inclined surfaces and demi-pointe position). Overall there was large variation in the types of measurement used to quantify postural control. Data were pooled where three or more individual studies measured similar constructs of postural control. Where measurement scales were inconsistent within these constructs, SMD (with 95% CI) was calculated.

Data pooled from 10 studies^{7,12,24,45,47–52} (Table 4) showed that PS-D (flat foot standing) was greater in subjects with FAI compared to healthy control subjects (SMD = 0.6 (95% CI: 0.2–1.0, $p = 0.002$) (Fig. 4). Just over half of these studies (six of ten) achieved 60% or more of the required quality criteria (Table 4). Four studies performed between limb comparisons in subjects with unilateral FAI^{12,14,24,53} (Table 4), and data could be pooled from three of these^{12,14,24} showing no difference between limbs (SMD = 0.2 (95% CI: –0.3 to 0.7, $p = 0.53$). Two studies, both of which were of low quality only assessed single-leg PS-D under challenging conditions.^{6,54} There was no difference in PS-D between subjects with FAI compared to controls standing on a translating platform,⁵⁴ and although data could not be extracted from one of the studies⁶ increased PS-D was reported (Table 4).

Table 3

Differences in joint angle position sense (JPS) between unstable and healthy ankles from individual studies.

Movement	Study	Direction of position sense (error °)	Comparison	Mean difference (95% CI) (or reported finding*)	% of QC met
Passive	36	PF	BG	0.4 (0.1–0.6)	69
	35	Inv.	BG	0.8 (–1.6 to 3.1)	63
	37	Combined for inv./ev.	BG BL	0.6 (–0.7 to 1.9) 0.3 (–1.2 to 1.7)	50
	34	Inv. (in PF)	BG	1.0 (0.2–1.9)	56
	39	Inv. (in 15° PF)	BG	0 (–2.3 to 2.3)	75
	38	Combined for inv./ev.	BG BL	1.6 (0.6–2.6) 1.1 (0.1–2.1)	31
Active	40	Slope box test (combined)	BG BL	0.6 (–1.8 to 3.0) 0.5 (–1.9 to 2.9)	94
	46	Combined for inv./ev.	BG	*No correlation between JPS error and functional ankle instability ($p < 0.05$)	63
	45	Inv.	BG	0.2 (–2.1 to 2.5)	69
	37	Combined for inv./ev.	BG BL	–0.8 (–1.7 to 1.0) –0.2 (–1.2 to 0.9)	50
	44	Inv.	BG BL	0.8 (0.6–1.0) 0.5 (0.3–0.7)	56
	34	Inv. (in PF)	BG	2.1 (0.4–3.7)	56
	39	Inv. (in PF 15°)	BG	0.4 (–1.3 to 2.1)	75
	43	Inv.	BG	0.9 (0.3–1.4)	31
	41	Inv. collapsed for gender	BG	0.8 (–0.3 to 2.3)	50
	38	Combined for inv./ev./neutral	BG BL	1.2 (0.2–2.3) 0.9 (–0.3 to 2.0)	31
	42	Inv.	BG	0.1 (–0.5 to 0.8)	31

QC = quality criteria; BG = between group; BL = between limb; PF = plantar flexion; inv. = inversion; ev. = eversion; DF = dorsiflexion

* Original finding reported as appropriate data could not be extracted to estimate effect.

Table 4

Differences for postural control variables between unstable and healthy ankles from individual studies.

Postural control measure	Study	Outcome used in analysis	Comparison	SMD, mean difference# 95% CI) (or reported finding*)	% of QC met
PS-D (single-leg stance)	7	Area of confidence ellipse	BG	1.1 (0.3 to 1.8)	31
	47	Area of confidence ellipse	BG	0.3 (−0.2 to 0.8)#	56
	53	Area of confidence ellipse	BG	1.0 (0.2 to 1/6)	50
			BL	0.3 (−0.5 to 1.0)	
	8	Area of confidence ellipse	BG	1.0 (0.2 to 1.9)	44
	52	Area of confidence ellipse	BG	0.8 (0.2 to 1.4)	56
	50	M-L sway distance	BG	0.2 (−0.6 to 0.9)	63
	49	Total sway (cm) (static, EO)	BG	0.7 (0.0 to 1.3)	69
	24	Transverse sway (mm)	BG	2.3 (1.4 to 3.2)	31
			BL	*no difference ($p > 0.05$)	
	48	Lateral sway (mm) (foot flat, static)	BG	−0.2 (−0.9 to 0.4)	69
	45	Transverse sway displacement (cm) (EC)	BG	0.6 (−0.7 to 1.9)	69
	14	M-L sway force (% body weight)	BL	0 (−1.0 to 0.9)	46
	12	Sway index (stable, EO)	BG	0.8 (−0.2 to 1.7)	69
			BL	0.2 (−0.7 to 1.1)	
	51	Stability Index level 6 (stable) (Biodex)	BG	0.2 (−0.6 to 0.9)	63
	6	Stabilometry (mean sway) on disc (without tape, before practice)	BG	*Increased in U subjects ($p < 0.01$)	25
	54	“ankle strategy” (index of postural PS) after medium M-L translation (Equitest)	BG	−0.2 (−1.2 to 0.7)	38
SEBT	57	Reach distance (cm)	BG	0.4 (−0.2 to 1.1)†	69
			BL	0.3 (−0.4 to 0.9)†	
	49	Reach distance (cm x height)	BG	0.5 (−0.1 to 1.2)†	69
	56	Reach distance (normalised for leg length)	BG	0.3 (−0.1 to 0.8)†	88
			BL	0.3 (−0.1 to 0.7)†	
TTS	55	Reach distance (% leg length)	BG	0.2 (−0.5 to 1.0)†	56
			BL	0.2 (−0.5 to 1.0)†	
	35	Single-leg jump (ms)			63
		M-L	BG	0.4 (−0.3 to 1.1)#	
		A-P		0.9 (0.6–1.3)#	
	60	Single-leg jump (ms)	BG	Unable to extract data^	56
	58	Single-leg jump (ms)			63
		M-L	BG	0.7 (0.3 to 1.0)#	
		A-P		0.6 (0.3–0.9)#	
	50	Single-leg jump (ms)			63
Other		M-L	BG	0.5 (0.01 to 0.9)#	
		A-P		0.9 (0.5 to 1.4)#	
	59	Single-leg jump (ms)			63
		M-L	BG	0.7 (0.1 to 1.3)#	
		A-P		0.4 (−0.03 to 0.8)#	
		BESS (error score)			
	63	Combined score 6 conditions	BG	1.0 (0.5 to 1.6)	69
	43	On foam (EO + EC)	BG	0.7 (−0.1 to 1.4)	31
	61	Standing balance time (s)	BG	−4.9 (−5.9 to −4.0)	56
	4	Modified Romberg (EO)	BL	Descriptive only	62
	62	Uniaxial balance evaluator—time spent out of balance (s)	BL	0.8 (0.3 to 1.2)	63
	36	Mean sway (°) level 6 (sensory organization test Equitest) tandem stance	BG	0.6 (−0.1 to 1.2)	69

#Mean difference estimates reported; *original finding reported as appropriate data could not be extracted to estimate effect; ^data collapsed between groups in original study as no difference reported; † positive values favor healthy ankles.

SMD= Standardised mean difference (SMD) estimates reported where measurement scales from individual studies were inconsistent); QC= quality criteria; PS-D= postural sway displacement; SEBT= Star Excursion Balance Test; TTS= time to stabilisation; EO= eyes open; EC= eyes closed; M-L: medio-lateral; A-P antero-posterior; BESS= Balance Error Scoring System; U= unstable.

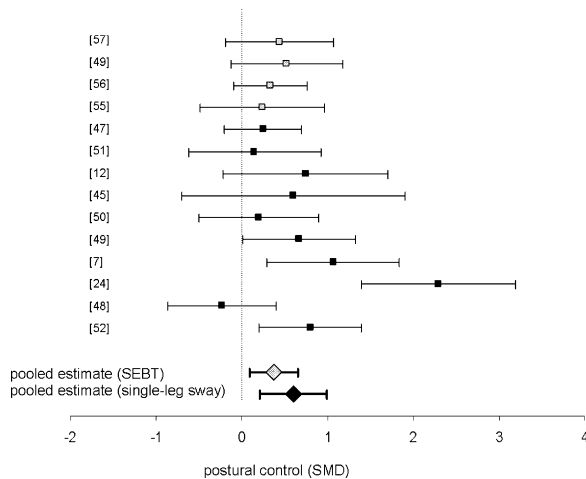


Fig. 4. Postural control expressed as standardised mean differences (SMD). Distribution of calculated estimates from 4 studies for star excursion balance test (SEBT) (■) and 10 studies on postural sway in single-leg stance (■) for differences between subjects with unstable ankles compared to healthy control subjects. Positive values for SEBT favor healthy control ankles. Pooled estimates (SMD, random effects model) for SEBT (◆) based on all 4 studies and postural sway (◆) based on all 10 studies. Horizontal bars represent 95% confidence intervals.

Data pooled from four studies,^{49,55–57} (Table 4) showed that reach distances were greater for controls compared to subjects with FAI (SMD = 0.4, 95% CI: 0.1–0.7, $p = 0.009$) (Fig. 4). All but one study⁵⁵ achieved 60% or more of the methodological quality criteria. There was no difference in reach distances between the unaffected and unstable limb in subjects with unilateral instability (SMD = 0.3, 95% CI: –0.1 to 0.6, $p = 0.1$) based on the pooling of data from three studies^{55–57} (Table 4).

The five studies investigating TTS from a single-leg jump^{35,50,58–60} (Table 4) reported medio-lateral (M-L) and antero-posterior (A-P) stabilisation times (ms). Data was pooled for four studies and these met greater than 60% of the methodological quality criteria^{35,50,58,59} (Fig. 5). Time to stabilisation was significantly faster in healthy control ankles compared to subjects with FAI for M-L (MD = 0.6 ms, 95% CI: 0.4–0.8, $p < 0.0001$) and A-P directions (MD = 0.7 ms, 95% CI: 0.4–1.0, $p < 0.0001$).

Less common measures of postural control included timed balance,^{61,62} BESS,^{43,63} subjective observation and perception of balance control,⁴ and tandem stance sway³⁶ (Table 4). These measures demonstrated inconsistent findings for sensorimotor deficit in unstable ankles. The two studies using the BESS^{43,63} yielded conflicting results, however the study by Docherty et al.⁶³ demonstrated balance deficits were of higher methodological quality (69% of quality criteria met compared to 31%⁴³).

4. Discussion

This systematic review is the first to independently pool data from individual studies on peroneal muscle reaction

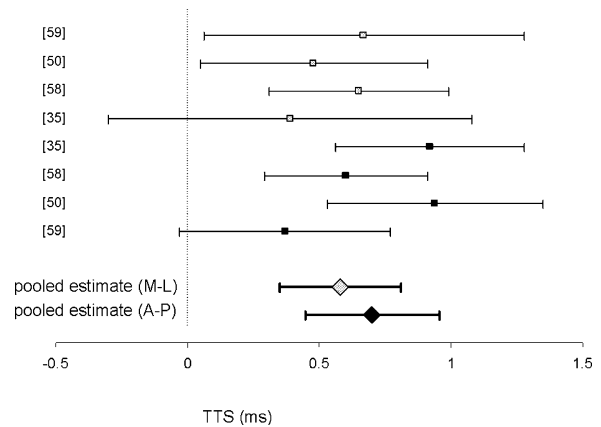


Fig. 5. Postural control—time to stabilisation (TTS). Distribution of calculated estimates for TTS from 4 studies for medio-lateral (M-L) sway (■) and antero-posterior (A-P) sway (■) for differences between subjects with unstable ankles compared to healthy control subjects. Positive values favor healthy control ankles. Pooled estimates (random effects model) for TTS based on all 4 studies for M-L (◆) and A-P (◆) sway. Horizontal bars represent 95% confidence intervals.

time, postural sway and JPS in a meta-analysis to determine the evidence for sensorimotor deficits in ankle instability. Results identified sensorimotor deficits for JPS and postural control in people with FAI.

Findings from individual studies on sensorimotor measures in FAI have been inconsistent to date, therefore pooling of data provided greater statistical power allowing a more precise estimation of differences between unstable and stable ankles for selected sensorimotor factors. There was lack of evidence for deficits of peroneal muscle reaction to perturbation. The data obtained for kinesthesia were inconclusive, however it appears that for movements in the frontal plane kinesthesia may be decreased.

Previous narrative reviews have suggested that traditional measures of single-leg sway under stable conditions may fail to elicit postural control deficits because they are not as challenging as dynamic conditions.⁶⁴ This may be a result of differing demands placed on mechanoreceptors under different conditions. Dynamic tests are likely to stimulate fast adapting mechanoreceptors compared to slow adapting receptors with static or slow moving tasks.⁶⁵ The results of this review demonstrate that postural control in single-leg stance is impaired in subjects with FAI compared to healthy controls. For measures of postural sway displacement there was a medium effect⁶⁶ (SMD = 0.6) for the difference between subjects with unstable ankles compared to healthy controls, and a small effect⁶⁶ for SEBT measures (SMD = 0.4). This deficit was not apparent when between limb comparisons were made in subjects with unilateral FAI.

Passive and active ankle JPS in subjects with FAI was decreased by 0.7° and 0.6° respectively when compared to healthy control subjects. While a sensorimotor deficit here is evident, movements with JPS testing occur relatively slowly, and it has been suggested that testing at slow speeds may

not be meaningful in relation to the mechanisms associated with ankle sprain.⁶⁵ Both JPS and kinesthesia are typically tested at slow speeds of around 0.5–2°/s with these measures targeting slow adapting mechanoreceptors.⁶⁷ Usually ankle sprains occur under forced rapid inversion movements so the relevance of such measurements to functional conditions of instability is uncertain.

For PMD measures it was not appropriate to pool data because of an inadequate number of studies with homogeneous measures. Only two studies performed between group comparisons; one high quality study (75% of methodological criteria met)¹¹ found an average 0.6° of movement detection deficit in the frontal plane. A lower quality study (56% of methodological criteria met) by the same authors³² investigating sagittal plane movement, observed no kinesthetic deficit. While it is possible that differing results may in part be attributed to biases associated with study quality, it may also be because of the different planes of movement. It is thought that for measures of proprioception such as kinesthesia, specific planes of movement are affected (for review see Refshauge⁶⁸). Also, some of the ambiguity between different sensorimotor measures may be because of differences in processing of information. Early theories suggest that signals for different aspects of proprioception such as JPS and kinesthesia are processed differently; separate lines of information can arise in muscle to signal movement and position so one aspect of proprioceptive acuity may be affected and not others.⁶⁹ Ambiguities in measures of sensory motor deficit may be because deficits are in fact not consistent across movement planes or because of differences in processing of signals for different aspects of sensorimotor control.

Peroneal reaction time was not impaired in subjects with FAI. Previous research has been conflicting, and it has been suggested that differences may be due to varied inclusion criteria for ankle instability.⁷⁰ This current review pooled studies with a broad definitional criterion for ankle instability (see Section 2) and showed that peroneal reaction time was not impaired. It has been proposed that measures of reactive muscle activity are probably unlikely to identify impairment because it is more likely that preparatory muscle activity is required to prevent ankle sprain and ankle giving way.^{65,70} It has been estimated that forced inversion from a standing position would put the lateral ligament complex under risk of damage after approximately 100 ms.⁷¹ Given that estimates of peroneal reaction time are around 85 ms, and that a further 90 ms elapses before contractile forces reach even half maximal voluntary contraction,^{72,73} it is unlikely that peroneal muscle reaction to inversion would be adequate to prevent injury to the lateral ligament complex. For this reason it is unlikely that a deficit in peroneal reaction time would be a key factor associated with ankle instability. It may be more important to evaluate sensorimotor function for preparatory muscle activity in unstable ankles due to the extremely short time periods to react to forces during functional tasks.⁶⁵

Many studies²³ included in this review performed between limb comparisons in subjects with unilateral FAI thereby using the contralateral ankle as a control. A possible advantage of this design is that potential unknown confounders for which control subjects are not matched may be accounted for. Designs comparing limbs of an individual assume that limb dominance does not affect the sensorimotor measure and that any possible sensorimotor deficit does not result from or contribute to central programming (that would thereby impact bilaterally).⁶⁵ Between limb comparisons for sensorimotor deficits have been criticized as evidence suggest that deficits can in fact be bilateral despite unilateral injury.^{8,53} For statistical analyses it is assumed that each data point represents an independent observation,⁷⁴ however, for between limb comparisons, each ankle is attached to the same person and this assumption is breached.⁷⁵ Because of within individual correlations, results for between limb comparisons may be biased.⁷⁶ Despite the shortcomings of between limb comparisons for sensorimotor deficits in subjects with unilateral ankle instability, a secondary analysis of sensorimotor measures for between limb differences in subjects with unilaterally instability was performed, as this was a common design in the literature reviewed. Where data could be pooled, JPS deficits existed which were consistent with findings for comparisons between unstable subjects and healthy controls. For postural sway however, deficits were not evident for between limb comparisons and this was inconsistent with findings for comparisons between unstable ankles and healthy control subjects. Interpretation of between limb results should be considered with caution because of potential statistical biases or the possibility that sensory motor deficits may occur bilaterally, despite unilateral injury, due to central processing of motor control information.

While the majority of studies clearly described measurement procedures, few provided information regarding measurement validity and reliability of these measures (methodological quality criterion item 20, Table S2). Most studies performed appropriate statistical analysis according to the criterion for item 18 (Supplementary data), however several studies performed multiple statistical comparisons and the interpretation of their findings may have been subject to type 1 error (e.g. Willems et al.³⁹).

Quality measurement tools designed specifically for observational studies that do not assess the effect of an intervention are limited. The original quality assessment tool¹⁸ was modified for use in this review as many of the criteria were not applicable as they related to items associated with potential design biases regarding effects of intervention. The modified rating tool used here is very similar to that used previously by Irving et al.¹⁹ in a review of factors associated with heel pain.

Identifying deficits in sensorimotor control associated with ankle instability can help to target intervention strategies towards these impairments for rehabilitation. In this review clear deficits in postural control and JPS were identified. The functional relevance of JPS deficits to instability

has been questioned because of the relatively slow testing speeds comparative to ankle injury,⁶⁵ and to our knowledge we know of no intervention studies specifically targeting this deficit which have been shown to be effective. For postural control there is evidence that strategies training balance and coordination are effective for functional outcomes such as preventing recurrent sprain and reducing symptoms of instability.^{77,78} Interestingly however, a recent systematic review found that despite improvements in functional outcomes, measures of postural sway did not improve following exercise training that included balance and motor coordination in subjects with functional ankle instability.⁷⁷

One of the limitations of this review is that pooling of data for subjects with FAI does not take into account the potential variation in disability of individuals within or across studies. Currently there is a lack of a criterion standard for measuring FAI and the broad levels of disability encompassed by the current definition.⁷⁹ As can be seen in Table S1, definitional criteria for functional instability from individual studies were broad, as were the criteria for this review and, as has been previously discussed, such broad criteria could potentially group subjects with FAI even though their symptoms differ vastly.⁸⁰ It may be that with discrete definitional criteria more homogeneous subgroups of FAI could be identified, and this may influence whether or not sensorimotor impairments are detected.

To account for new research that became available prior to the publication of this study, an additional electronic search was performed in November 2008. A further 16 papers were identified, of which six reported relevant data that met the original inclusion criteria, and where possible, effect sizes were recalculated. Four papers examined postural control^{81–84} with this data adding strength to the original finding that postural sway is impaired in single-leg stance (0.6, 95% CI: 0.3–0.9; $p < 0.0001$), and during SEBT,⁸² (0.4, 95% CI: 0.1–0.6; $p = 0.008$). A further two studies were identified for AJPS.^{83,85} Similarly, these data increase support for the original finding that AJPS is impaired in subjects with FAI (0.7°, 95% CI: 0.4° to 1.0°; $p < 0.0001$). The seventh study that met inclusion criteria assessed PMD in the frontal plane⁸⁶ and reported no deficit in FAI compared to healthy controls. This finding is in contrast to the only other study¹¹ identified in the original search for this sensorimotor outcome.

5. Conclusion

Findings from individual studies on sensorimotor factors associated with FAI commonly yield conflicting results. This systematic review pooled data to provide clearer evidence about deficits for several measures of sensorimotor control in subjects with ankle instability. For measures of postural sway in single-leg stance, dynamic balance (SEBT), TTS and JPS there is evidence that deficits exist in subjects with FAI compared to healthy controls. For peroneal muscle reaction to perturbation sensorimotor deficits were not evident.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.jsams.2009.03.004](https://doi.org/10.1016/j.jsams.2009.03.004).

References

1. Yeung MS, Chan KM, So CH, et al. An epidemiological survey on ankle sprain. *Br J Sports Med* 1994;**28**(2):112–6.
2. Beynon BD, Renstrom PA, Alosa DM, et al. Ankle ligament injury risk factors: a prospective study of college athletes. *J Orthop Res* 2001;**19**(2):213–20.
3. Hertel J. Functional anatomy, pathomechanics, and pathophysiology of lateral ankle instability. *J Athl Train* 2002;**37**(4):364–75.
4. Lentell GL, Katzman LL, Walters MR. The relationship between muscle function and ankle stability. *J Orthop Sports Phys Ther* 1990;**11**(12):605–11.
5. Hertel J. Functional instability following lateral ankle sprain. *Sports Med* 2000;**29**(5):361–71.
6. Leanderson J, Ekstam S, Salomonsson C. Taping of the ankle—the effect on postural sway during perturbation, before and after a training session. *Knee Surg Sports Traumatol Arthrosc* 1996;**4**(1):53–6.
7. Tropp H, Odenrick P. Postural control in single-limb stance. *J Orthop Res* 1988;**6**(6):833–9.
8. Gauffin H, Tropp H, Odenrick P. Effect of ankle disk training on postural control in patients with functional instability of the ankle joint. *Int J Sports Med* 1988;**9**(2):141–4.
9. Lofvenberg R, Karrholm J, Sundelin G, et al. Prolonged reaction time in patients with chronic lateral instability of the ankle. *Am J Sports Med* 1995;**23**(4):414–7.
10. Garn SN, Newton RA. Kinesthetic awareness in subjects with multiple ankle sprains. *Phys Ther* 1988;**68**(11):1667–71.
11. Refshauge KM, Kilbreath SL, Raymond J. Deficits in detection of inversion and eversion movements among subjects with recurrent ankle sprains. *J Orthop Sports Phys Ther* 2003;**33**(4):166–73, discussion 173–6.
12. Bernier JN, Perrin DH, Rijke A. Effect of unilateral functional instability of the ankle on postural sway and inversion and eversion strength. *J Athl Train* 1997;**32**(3):226–32.
13. Ebig M, Lephart SM, Burdett RG, et al. The effect of sudden inversion stress on EMG activity of the peroneal and tibialis anterior muscles in the chronically unstable ankle. *J Orthop Sports Phys Ther* 1997;**26**(2):73–7.
14. Isakov E, Mizrahi J. Is balance impaired by recurrent sprained ankle? *Br J Sports Med* 1997;**31**(1):65–7.
15. Richie Jr DH. Functional instability of the ankle and the role of neuromuscular control: a comprehensive review. *J Foot Ankle Surg* 2001;**40**(4):240–51.
16. Herbert RD, Jamtvedt G, Mead J, et al. *Practical evidence-based physiotherapy*. London: Elsevier Butterworth Heinemann; 2005.

17. Smith GD, Egger M. Going beyond the grand mean: subgroup analysis of randomised trials. In: Egger M, Smith GD, Altman DG, editors. *Systematic reviews in health care: meta-analysis in context*. London: BMJ; 2001.
18. Downs SH, Black N. The feasibility of creating a check list for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. *J Epidemiol Commun Health* 1998;**52**(6):377–84.
19. Irving DB, Cook JL, Menz HB. Factors associated with chronic plantar heel pain: a systematic review. *J Sci Med Sport* 2006;**9**(1–2): 11–22.
20. The Cochrane Collaboration. *Review Manager (RevMan) [Computer program] Version 4.2 for Windows*. Copenhagen: The Nordic Cochrane Centre; 2003.
21. Khin Myo H, Ishii T, Sakane M, et al. Effect of anesthesia of the sinus tarsi on peroneal reaction time in patients with functional instability of the ankle. *Foot Ankle Int* 1999;**20**(9):554–9.
22. Shima N, Maeda A, Hirohashi K. Delayed latency of peroneal reflex to sudden inversion with ankle taping or bracing. *Int J Sports Med* 2005;**26**(6):476–80.
23. Vaes P, Van Gheluwe B, Duquet W. Control of acceleration during sudden ankle supination in people with unstable ankles. *J Orthop Sports Phys Ther* 2001;**31**(12):741–52.
24. Konradsen L, Ravn JB. Prolonged peroneal reaction time in ankle instability. *Int J Sports Med* 1991;**12**(3):290–2.
25. Vaes P, Duquet W, Van Gheluwe B. Peroneal reaction times and eversion motor response in healthy and unstable ankles. *J Athl Train* 2002;**37**(4):475–80.
26. Javed A, Walsh HPJ, Lees A. Peroneal reaction time in treated functional instability of the ankle. *Foot Ankle Surg* 1999;**5**(3):159–66.
27. Karlsson J, Andreasson GO. The effect of external ankle support in chronic lateral ankle joint instability. An electromyographic study. *Am J Sports Med* 1992;**20**(3):257–61.
28. Rosenbaum D, Becker HP, Gerngroß H, et al. Peroneal reaction times for diagnosis of functional ankle instability. *Foot Ankle Surg* 2000;**6**(1):31–8.
29. Schmidt R, Benesch S, Hald R, et al. Die inzidenz und wertigkeit des propriozeptiven defizites bei patienten mit chronischer instabilitaet des oberen sprunggelenkes. Incidence and relevance of proprioceptive deficits in patients with chronic ankle instability. *Dtsch Z Sportmed* 2005;**56**(11):378–82.
30. Forkin DM, Koczur C, Battle R, et al. Evaluation of kinesthetic deficits indicative of balance control in gymnasts with unilateral chronic ankle sprains. *J Orthop Sports Phys Ther* 1996;**23**(4):245–50.
31. Hubbard TJ, Kaminski TW. Kinesthesia is not affected by functional ankle instability status. *J Athl Train* 2002;**37**(4):481–6.
32. Refshauge KM, Kilbreath SL, Raymond J. The effect of recurrent ankle inversion sprain and taping on proprioception at the ankle. *Med Sci Sports Exerc* 2000;**32**(1):10–5.
33. Lentell G, Baas B, Lopez D, et al. The contributions of proprioceptive deficits, muscle function, and anatomic laxity to functional instability of the ankle. *J Orthop Sports Phys Ther* 1995;**21**(4):206–15.
34. Boyle J, Negus V. Joint position sense in the recurrently sprained ankle. *Aust J Physiother* 1998;**44**(3):159–63.
35. Brown C, Ross S, Mynark R, et al. Assessing functional ankle instability with joint position sense, time to stabilization, and electromyography. *J Sport Rehabil* 2004;**13**(2):122–34.
36. Fu AS, Hui-Chan CW. Ankle joint proprioception and postural control in basketball players with bilateral ankle sprains. *Am J Sports Med* 2005;**33**(8):1174–82.
37. Gross MT. Effects of recurrent lateral ankle sprains on active and passive judgements of joint position. *Phys Ther* 1987;**67**(10):1505–9.
38. Liu Y, Jeng S, Lee AJY. The influence of ankle sprains on proprioception. *J Exerc Sci Fitness* 2005;**3**(1):33–8.
39. Willems T, Witvrouw E, Verstuyft J, et al. Proprioception and muscle strength in subjects with a history of ankle sprains and chronic instability. *J Athl Train* 2002;**37**(4):487–93.
40. Halasi T, Kynsburg A, Tallay A, et al. Changes in joint position sense after surgically treated chronic lateral ankle instability. *Br J Sports Med* 2005;**39**(11):818–24.
41. Hartsell HD. The effects of external bracing on joint position sense awareness for the chronically unstable ankle. *J Sport Rehabil* 2000;**9**(4):279–89.
42. Jerosch J, Hoffstetter I, Bork H, et al. The influence of orthoses on the proprioception of the ankle joint. *Knee Surg Sports Traumatol Arthrosc* 1995;**3**(1):39–46.
43. Jerosch J, Bischof M. Proprioceptive capabilities of the ankle in stable and unstable joints. *Sports Exerc Inj* 1996;**2**(4):167–71.
44. Konradsen L, Magnusson P. Increased inversion angle replication error in functional ankle instability. *Knee Surg Sports Traumatol Arthrosc* 2000;**8**(4):246–51.
45. You SH, Granata KP, Bunker LK. Effects of circumferential ankle pressure on ankle proprioception, stiffness, and postural stability: a preliminary investigation. *J Orthop Sports Phys Ther* 2004;**34**(8):449–60.
46. Docherty CL, Arnold BL, Hurwitz S. Contralateral force sense deficits are related to the presence of functional ankle instability. *J Orthop Res* 2006;**24**(7):1412–9.
47. Baier M, Hopf T. Ankle orthoses effect on single-limb standing balance in athletes with functional ankle instability. *Arch Phys Med Rehabil* 1998;**79**(8):939–44.
48. Hiller CE, Refshauge KM, Beard DJ. Sensorimotor control is impaired in dancers with functional ankle instability. *Am J Sports Med* 2004;**32**(1):216–23.
49. Nakagawa L, Hoffman M. Performance in static, dynamic, and clinical tests of postural control in individuals with recurrent ankle sprains. *J Sport Rehabil* 2004;**13**(3):255–68.
50. Ross SE, Guskiewicz KM. Examination of static and dynamic postural stability in individuals with functionally stable and unstable ankles. *Clin J Sport Med* 2004;**14**(6):332–8.
51. Rozzi SL, Lephart SM, Sterner R, et al. Balance training for persons with functionally unstable ankles. *J Orthop Sports Phys Ther* 1999;**29**(8):478–86.
52. Tropp H, Odenrick P, Gillquist J. Stabilometry recordings in functional and mechanical instability of the ankle joint. *Int J Sports Med* 1985;**6**(3):180–2.
53. Tropp H. Pronator muscle weakness in functional instability of the ankle joint. *Int J Sports Med* 1986;**7**(5):291–4.
54. Pintsaar A, Brynhildsen J, Tropp H. Postural corrections after standardised perturbations of single limb stance: effect of training and orthotic devices in patients with ankle instability. *Br J Sports Med* 1996;**30**(2):151–5.
55. Gribble PA, Hertel J, Denegar CR, et al. The effects of fatigue and chronic ankle instability on dynamic postural control. *J Athl Train* 2004;**39**(4):321–9.
56. Hertel J, Braham RA, Hale SA, et al. Simplifying the star excursion balance test: analyses of subjects with and without chronic ankle instability. *J Orthop Sports Phys Ther* 2006;**36**(3):131–7.
57. Olmsted LC, Carcia CR, Hertel J, et al. Efficacy of the star excursion balance tests in detecting reach deficits in subjects with chronic ankle instability. *J Athl Train* 2002;**37**(4):501–6.
58. Ross SE, Guskiewicz KM. Effect of coordination training with and without stochastic resonance stimulation on dynamic postural stability of subjects with functional ankle instability and subjects with stable ankles. *Clin J Sport Med* 2006;**16**(4):323–8.
59. Ross SE, Guskiewicz KM, Yu B. Single-leg jump-landing stabilization times in subjects with functionally unstable ankles. *J Athl Train* 2005;**40**(4):298–304.
60. Wikstrom EA, Tillman MD, Borsa PA. Detection of dynamic stability deficits in subjects with functional ankle instability. *Med Sci Sports Exerc* 2005;**37**(2):169–75.
61. Chrintz H, Falster O, Roed J. Single-leg postural equilibrium test. *Scand J Med Sci Sports* 1991;**1**(4):244–6.
62. Ryan L. Mechanical stability, muscle strength and proprioception in the functionally unstable ankle. *Aust J Physiother* 1994;**40**(1):41–7.

63. Docherty CL, Valovich McLeod TC, Shultz SJ. Postural control deficits in participants with functional ankle instability as measured by the balance error scoring system. *Clin J Sport Med* 2006;**16**(3):203–8.
64. Riemann B, Caggiano N, Lephart S. Examination of a clinical method of assessing postural control during a functional performance task. *J Sport Rehabil* 1999;**8**(3):171–83.
65. Wikstrom EA, Tillman MD, Chmielewski TL, et al. Measurement and evaluation of dynamic joint stability of the knee and ankle after injury. *Sports Med* 2006;**36**(5):393–410.
66. Cohen J. *Statistical power analysis for the behavioral sciences*. 2nd ed. Hillsdale, New Jersey: Earlbaum; 1988.
67. Riemann B, Myers J, Lephart S. Sensorimotor system measurement techniques. *J Athl Train* 2002;**37**(1):85–98.
68. Refshauge KM. Proprioception and joint pathology. In: Gandevia SC, Proske U, Stuart DG, editors. *Sensorimotor control of movement and posture*. London: Kluwer Academic/Plenum Publishers; 2002.
69. McCloskey DI. Differences between the senses of movement and position shown by the effects of loading and vibration of muscles in man. *Brain Res* 1973;**61**:119–31.
70. Delahunt E. Peroneal reflex contribution to the development of functional instability of the ankle joint. *Phys Ther Sport* 2007;**8**(2):98–104.
71. Konradsen L, Voigt M, Højsgaard C. Ankle inversion injuries: the role of the dynamic defense mechanism. *Am J Sports Med* 1997;**25**(1):54–8.
72. Ashton-Miller JA, Ottaviani RA, Hutchinson C, et al. What best protects the inverted weight bearing ankle against further inversion? Evertor muscle strength compares favorably with shoe height, athletic tape, and three orthoses. *Am J Sports Med* 1996;**24**(6):800–9.
73. Ottaviani R, Ashton-Miller J, Kothari S, et al. Basketball shoe height and the maximal muscular resistance to applied ankle inversion and eversion moments. *Am J Sports Med* 1995;**23**(4):418–23.
74. Bland JM, Altman DG. Statistical notes: correlation, regression and repeated data. *Br Med J* 1994;**308**:896.
75. Menz HB. Two feet or one person? Problems associated with statistical analysis of paired data in foot and ankle medicine. *Foot* 2004;**14**:2–5.
76. Bryant D, Havey TC, Roberts R, et al. How many patients? How many limbs? Analysis of patients or limbs in the orthopaedic literature: a systematic review. *J Bone Joint Surg* 2006;**88A**(1):41–5.
77. van der Wees PJ, Lenssen AF, Hendricks EJM, et al. Effectiveness of exercise therapy and manual mobilisation in acute ankle sprain and functional instability: a systematic review. *Aust J Physiother* 2006;**52**(1):27–37.
78. Verhagen E, van der Beek A, Twisk J, et al. The effect of a proprioceptive balance board training program for the prevention of ankle sprains: A prospective controlled trial. *Am J Sports Med* 2004;**32**(6):1385–93.
79. Hiller CE, Refshauge KM, Bundy AC, et al. The Cumberland ankle instability tool: a report of validity and reliability testing. *Arch Phys Med Rehabil* 2006;**87**(9):1235–41.
80. Hertel J, Kaminski TW. Second international ankle symposium summary statement. *J Orthop Sports Phys Ther* 2005;**35**(5):A–2–6.
81. Hertel J, Olmsted-Kramer LC. Deficits in time-to-boundary measures of postural control with chronic ankle instability. *Gait Posture* 2007;**25**(1):33–9.
82. Hale SA, Hertel J, Olmsted-Kramer LC. The effect of a 4-week comprehensive rehabilitation program on postural control and lower extremity function in individuals with chronic ankle instability. *J Orthop Sports Phys Ther* 2007;**37**(6):303–11.
83. Santos MJ, Liu W. Possible factors related to functional ankle instability. *J Orthop Sports Phys Ther* 2008;**38**(3):150–7.
84. Hiller CE, Refshauge KM, Herbert RD, et al. Balance and recovery from a perturbation are impaired in people with functional ankle instability. *Clin J Sport Med* 2007;**17**(4):269–75.
85. Nakasa T, Fukuhara K, Adachi N, et al. The deficit of joint position sense in the chronic unstable ankle as measured by inversion angle replication error. *Arch Orthop Trauma Surg* 2008;**128**(5):445–9.
86. de Noronha M, Refshauge KM, Kilbreath SL, et al. Loss of proprioception or motor control is not related to functional ankle instability: an observational study. *Aust J Physiother* 2007;**53**(3):193–8.